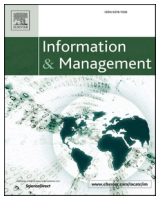
Information & Management 62 (2025) 104199



Contents lists available at ScienceDirect

Information & Management

journal homepage: www.elsevier.com/locate/im





Digitalization and financial performance: A supply chain perspective

Yinan Qi [a], Kexin Hou [a], Min Zhang [b], Tianlong Pu [c,*], Hangfei Guo [b]

[a] Business School, University of International Business and Economics, Beijing, PR China
[b] Queen's Business School, Queen's University Belfast, Belfast, UK
[c] Shenzhen Institute for Advanced Study, University of Electronic Science and Technology of China, Shenzhen, PR China

ARTICLE INFO

Keywords:
Digitalization
Digitalization proximity
Financial performance
Supply chain

ABSTRACT

Digitalization is widely regarded as a strategic approach to gaining competitive advantage. However, some firms have suffered from the digitalization paradox, meaning that investments in digitalization cannot always achieve the expected performance improvement. This study examines how the digitalization of a firm affects its financial performance and the moderating effect of digitalization proximity between the firm and its supplier. The results suggest an inverted U-shaped relationship between a firm's digitalization and financial performance. Furthermore, the positive moderating role of digitalization proximity in this nonlinear relationship suggests that alignment with suppliers in digitalization allows a firm to materialize the benefits of digitalization. This study provides evidence of the complex relationship between digitalization and financial performance. It empirically demonstrates the importance of aligning with supply chain partners for digitalization to enhance performance benefits. It also offers practical guidance for managers in navigating the evolving digital landscape.

1. Introduction

Digitalization involves investing in information technologies, assets, and tools to transform business processes and activities [1–3]. In this study, we focus on the digital technologies that drive business transformation, leverage the value of data, facilitate the decision-making process, and create new market opportunities [4–6]. Specifically, we concentrate on artificial intelligence (AI), blockchain, cloud computing, big data analytics, and digital applications, which are widely adopted and extensively studied and have generated significant economic value (e.g., [7,8]). AI technology can assist in identifying patterns, enabling automated and intelligent production and operation processes, and providing scientifically validated recommendations [9]. Big data analytics and cloud computing technology offer enhanced capabilities for real-time response, analysis, and monitoring [10,11]. Blockchain technology disrupts traditional transaction models, enabling trust-free economic transactions [12]. Digital application technologies translate foundational technologies into practical solutions, lowering the barrier to entry and enabling businesses to harness the benefits of digital innovation swiftly [13]. These technologies help firms to invent and produce new products or processes and enable ongoing technical improvement and innovation [6,14]. Researchers argue that digitalization is critical for firms to improve performance outcomes [15]. However, despite substantial investments in digitalization, firms may not achieve the expected performance enhancements, a phenomenon known as the digitalization paradox [5,16]. For example, research has shown that only 17 % of Chinese firms have a digital edge that translates into improved financial performance, and the average increase is approximately 11 % [17].

The literature shows inconsistent results regarding how digitalization influences firm performance. Some studies find that digitalization is positively associated with performance outcomes (e.g., [11,18–20]), whereas others argue that addressing various challenges is necessary to achieve performance improvement through digitalization (e.g., [16,21,22]). Recent research has revealed inverted U-shaped relationships between digitalization and innovation performance [23], profitability [24], environmental information disclosure [1], and environmental performance [25]. Studies have also reported mixed findings on the impact of digitalization on financial performance (Appendix 1), highlighting the need for further empirical research. For example, Kohtamäki et al. [26] identified an inverted U-shaped relationship based on a questionnaire survey of Swedish manufacturing firms. In contrast, Bhandari et al. [27] found a U-shaped relationship based on a sample of US-listed manufacturing firms. Therefore, our first research question is

* Corresponding author.
E-mail addresses: yinanqi@uibe.edu.cn (Y. Qi), 202200320080@uibe.edu.cn (K. Hou), M.Zhang@qub.ac.uk (M. Zhang), Tianl0115@163.com (T. Pu), h.guo@qub.ac.uk (H. Guo).

https://doi.org/10.1016/j.im.2025.104199
Received 8 April 2024; Received in revised form 3 June 2025; Accepted 10 June 2025
Available online 10 June 2025
0378-7206/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

the following: How does digitalization affect a firm's financial performance?

Digitalization has evolved from an internal focus to a supply chain perspective [28,29]. Previous research has highlighted the significance of supply chain digital collaboration and the alignment of digitalization among supply chain partners in achieving competitive advantages [30,31]. Digitalization proximity refers to the alignment of digitalization between a firm and its supplier [32,33]. It denotes the similarities in investments in specific digital technologies between a firm and its supplier. Thus, increases in digitalization proximity are expected to facilitate supply chain digital collaboration between firms and their suppliers [34]. However, there is limited empirical evidence on the performance implications of the alignment of firms and their suppliers in digitalization [29]. Therefore, our second research question is as follows: How does digitalization proximity affect the impact of digitalization on financial performance?

We employ a two-way fixed effects model complemented by robustness checks to investigate the research questions. We construct a panel data set encompassing 788 listed Chinese manufacturing firms and their 1448 suppliers from 2010 to 2020. We find that an inverted U-shaped impact of digitalization on financial performance is well supported at the firm level, with the turning point located at the extreme end of our samples. The results further reveal a positive moderating role of digitalization proximity in the digitalization–financial performance linkage.

This study contributes to the literature by clarifying the complex relationship between digitalization and financial performance. The result is consistent with Kohtamäki et al. [26], providing further empirical evidence of the inverted U-shaped relationship between digitalization and financial performance. In addition, it corroborates the existence of the digitalization paradox (e.g., [20,23]). It also investigates digitalization within a supply chain context. A firm can derive greater financial benefits from digitalization when its efforts align with its suppliers. This finding not only aligns with previous studies highlighting the beneficial effects of technology proximity (e.g., [33]) but also highlights the pivotal role of supply chains in determining the success of a firm's digitalization strategy (e.g., [29,35]). Our findings remind managers to exercise caution, avoiding indiscriminate increases in digitalization investments. It is also imperative for managers to recognize that aligning digitalization efforts is instrumental in reaping the comprehensive benefits.

2. Literature review and hypotheses development

2.1. Digitalization and financial performance

Digitalization is the process by which firms integrate and utilize advanced digital technologies to optimize operational processes, improve supply chain management, and ultimately reach strategic goals and achieve competitive advantages [28,36]. It allows firms to embrace new approaches in management, offering novel avenues for value creation [37]. Proponents emphasize that implementing digitalization enhances operational performance [19,38], innovation performance [39–41], environmental performance and sustainability [42,43], and productivity [44].

Researchers have adopted different approaches, such as questionnaire surveys [11,26,45–48], text mining [27,49], and field experiments [50], to measure and empirically investigate the impact of digitalization and report a positive association between digitalization and financial performance. For example, based on a survey of U.S. and European manufacturers, Abou-Foul et al. [46] showed that digitalization has a direct and positive effect on a firm's financial performance. Using text mining and samples from Chinese listed companies, researchers find that digitalization can significantly improve stock liquidity and firm performance [49]. Moreover, studies have shown that adopting data-driven decision-making can improve firm productivity [51] and help small and medium-sized enterprises establish a competitive advantage [50]. However, achieving financial success through digitalization remains challenging and risky [15,37]. An increasing body of research has revealed paradoxes in the impact of digitalization on firm performance, such as the paradox of asymmetrical costs and benefits [52] and the paradox of flexibility and stability of the organizational structure [53]. Researchers have also examined the nonlinear relationship between digitalization and firm performance. For example, Kohtamäki et al. [26] reported an inverted U-shaped relationship between digitalization and financial performance, which is moderated by servitization. Brynjolfsson et al. [54] argued that there is a J-curve between investment in AI and productivity growth. Li [25] identified an inverted U-shaped relationship between digitalization and a firm's environmental performance. Gong et al. [23] found an inverted U-shaped relationship between digitalization and radical innovation performance. Xiang and Jiang [24] found that digitalization first decreases and then increases the performance of commercial banks. Bhandari et al. [27] confirmed that digitalization has a U-shaped link with a firm's financial performance. Hence, ongoing investment in digitalization may not necessarily result in the anticipated revenue growth.

We argue that after the initial boost to financial performance, the positive effect of digitalization investment gradually decreases until it flips over to a negative effect. Specifically, digitalization enhances firms' information processing capacity [13]. It assists firms in broadening their knowledge boundaries, allowing them to leverage a broad spectrum of informational resources [55]. This knowledge can be used to optimize production processes and improve operational efficiency [39]. However, the marginal benefits of digitalization are diminishing, as digital technologies may yield redundant information. A firm may also lack an appropriate management system to process and apply the vast amount of data acquired from advanced digital technologies to improve operational efficiency. Hence, a sustained increase in digitalization will result in diminishing returns on investment [56]. Continuous digital investment crowds out resources to develop capabilities that improve operational efficiency, such as business process reengineering and employee training [57]. Empirical evidence shows that operational efficiency is associated positively with financial performance [58–60]. Thus, the improvement in financial performance diminishes as digitalization continues to increase.

Moreover, the costs associated with digitalization are expected to increase, negatively affecting operational efficiency and hence financial performance. As digitalization progresses, the demand for various digital technologies increases, requiring firms to acquire new capabilities to drive business development. For example, digitalization increases data collection, storage, and management costs. In addition to acquisition costs, digitalization requires firms to upgrade their operational processes and systems [23]. Indeed, during digitalization, firms must reassess their strategy, organizational structure and business ecosystem [45], increasing hidden management costs. Firms must employ and train specialized personnel to implement digital technologies fully, increasing human resource costs [61]. Hence, the effect of digitalization on financial performance involves diminishing marginal benefits and increasing costs. Therefore, we propose the following hypothesis:

H1. Digitalization has an inverted U-shaped relationship with financial performance.

2.2. The moderating role of digitalization proximity

Recent studies have viewed digitalization as a strategic initiative

involving supply chain members [15,28,29,40]. The effectiveness of the digitalization of firms can be influenced by their suppliers' digital strategies and capabilities [62]. Data-driven collaboration in supply chains helps reduce information asymmetry and coordination barriers, enhancing the efficiency and performance of the supply chain [35,50]. However, due to technological and financial constraints, the extent to which firms have adopted digital technologies varies considerably [63]. Such variations may result in supply chain members adopting different digital technologies [29]. This misalignment increases the difficulty of collaboration between supply chain members. Thus, it may undermine the efficiency and effectiveness of collaboration among supply chain partners. When a firm invests in digitalization, considering its suppliers' digitalization strategies is crucial.

Digitalization proximity captures the alignment in digitalization between firms and their suppliers and reflects the closeness in their investment choices regarding digital technologies [32,33]. It is related to the strategic focus and the allocation of investments in digital technologies [64]. Digital technologies empower supply chain members to engage in real-time information sharing, collaborative planning, and enhanced responsiveness, thus strengthening supply chain transparency, flexibility, and agility [8]. However, digital technologies have diverse attributes, functions, and applications, making them difficult to integrate. Effective collaboration cannot be achieved if a firm and its suppliers apply different technologies and tools during digitalization.

We argue that digitalization proximity between firms and their suppliers enhances the impact of digitalization on financial performance in two ways. First, it reduces transaction costs. The proximity of digitalization between a firm and its suppliers breaks "data silos," enhancing information transparency and reducing costs of communication and information searching [41]. A firm that maintains close digital ties with its suppliers is more likely to alleviate coordination barriers in supply chain processes and reduce the likelihood of suppliers' opportunistic behavior [33], thereby reducing default costs. Second, digitalization proximity improves coordination benefits. Digitalization proximity establishes an accurate and rapid data exchange channel between a firm and its suppliers, enhancing the accuracy and efficiency of the firm's transactions [65]. This alignment also promotes consistency among organizational strategies, production, and operations within the supply chain through standardized data sharing and interoperability [41]. Hence, a firm can benefit from streamlined transactions caused by greater digitalization proximity. Therefore, as digitalization proximity increases, the benefit of digitalization is strengthened and its cost is reduced, leading to a steepening of the inverted U-shaped curve between digitalization and financial performance [66]. Hence, high digitalization proximity implies that a firm can benefit more from digitalization investments, causing the inflexion point of the nonlinear curve to shift rightward. Therefore, we propose the following hypothesis:

H2. Digitalization proximity moderates the inverted U-shaped relationship between digitalization and financial performance positively.

Fig. 1 shows the theoretical model.

3. Methodology

3.1. Sample and data

We select a sample of listed manufacturing firms from the China Stock Market and Accounting Research (CSMAR) database from 2012 to 2023. Then we collect data on each firm's supply chain relationship from the FactSet database, which provides transaction information between a firm and its supplier based on the actual purchase [67]. If a purchasing transaction occurs between a firm and a supplier in a year, we view this as a supply chain relationship. We construct a panel dataset of listed Chinese manufacturing firms paired with their suppliers. Firms with missing values and labeled special treatment are removed. The final sample comprises 4472 firm–supplier–year observations, including 788 firms and 1448 suppliers over 12 years. The analysis is based on the firm–supplier dyad because digitalization proximity is defined at the individual supplier relationship level. The firm–supplier dyad also allows us to preserve the supplier's idiosyncratic effects.

3.2. Variables

3.2.1. Digitalization

Following Tao et al. [68], we measure the level of a firm's digitalization using the ratio of the firm's digitalized intangible assets and digitalized construction in progress to total intangible assets and total construction in progress. We construct a feature lexicon based on the literature and research reports (Appendix 2). We classify all feature words into five categories: AI, blockchain, cloud computing, big data analytics, and digital applications [13,69]. We subsequently screen the intangible assets and construction-in-progress items from the firm's notes to the annual financial report manually. We set them as digitalized intangible assets or digitalized construction in progress if they contain the keywords. Digitalization (*DIGIT*) is computed as

$$DIGIT_{i,t} = \frac{DIGINTAN_{i,t} + DIGCIP_{i,t}}{INTAN_{i,t} + CIP_{i,t}}, \quad (1)$$

where $DIGINTAN_{i,t}$ denotes firm i's digitalized intangible assets at the end of period t and $DIGCIP_{i,t}$ denotes firm i's digitalized construction in progress at the end of period t. $INTAN_{i,t}$ and $CIP_{i,t}$ denote the total intangible assets and total construction in progress of firm i at the end of period t.

3.2.2. Financial performance

We assess a firm's financial performance using its annual return on equity (ROE), computed as the ratio of net profit to net assets.

3.2.3. Digitalization proximity

Following Benner and Waldfogel [32], we use the Euclidean distance

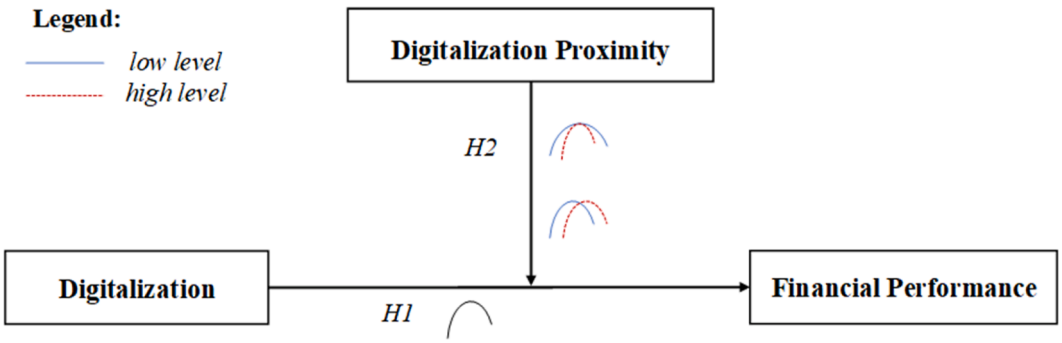


Fig. 1. Theoretical model.

to measure digitalization proximity (*PROXIMITY*). This approach can capture the similarity between a firm and its supplier in terms of digitalization [70]. The measure is calculated via the equation

$$PROXIMITY_{i,j,t} = -1 * \sqrt{\sum_{k=1}^{5} \left(d_{ikt} - d_{jkt}\right)^2}, \quad (2)$$

where $PROXIMITY_{i,j,t}$ represents the similarity in digitalization between focal firm i and its supplier j during period t. We classify investment in digital technologies into five categories: AI, cloud computing, blockchain, big data, and digital applications. d_{ikt} denotes the ratio of intangible assets and construction in progress related to technology in category k to the total amount of intangible assets and construction in progress at the end of period t for firm i. Similarly, d_{jkt} denotes the ratio of intangible assets and construction in progress related to technology in category k to the total amount of intangible assets and construction in progress at the end of period t for supplier j. The Euclidean distance between d_{it} and d_{jt} reflects the difference in digitalization between firm i and supplier j at the end of period t. We then take the negative transformation of the Euclidean distance to calculate the digitalization proximity.

3.2.4. Control variables

We incorporate several control variables to accommodate firm-level (F_{it}), supplier-level (S_{jt}), and firm–supplier dyad-level (D_{ijt}) heterogeneity. First, we control for variables that embody a firm's resources. Firm size ($SIZE^F$) is measured by the natural logarithm of the number of employees of a firm. Firm age (AGE^F) is measured as the logarithm of the number of years from the establishment year to the reporting year. Large and old firms tend to accumulate more knowledge and experience. We also control for the cash turnover ratio ($CASH^F$), which is determined by the ratio of total sales generated by operating activities to the average cash balance, and the sales growth rate ($SALEGR^F$), which is the change in total sales for the current period relative to the prior period [71]. The cash turnover ratio and sales growth rate capture the firm's financial resources.

Second, we control for a firm's financial risk, which influences financial performance. We use leverage (LEV^F), measured as the ratio of total liabilities to total assets, as a control variable for measuring the debt repayment capability of a firm [68]. Moreover, a firm's accounts payable turnover (PAY^F) is controlled to capture its bargaining power, computed by dividing net operating costs by the average accounts payable balance.

Third, a firm's capital market performance is also expected to affect its financial performance. We thus control for stock return volatility (SRV^F) and the shareholding ratio (SHR^F). The former is calculated as the annualized standard deviation of monthly stock returns, and the latter is measured as the proportion of shares held by the largest shareholder [72]. Furthermore, we include Tobin's Q (TQ^F), which captures corporate value in the stock market. It is calculated as the ratio of the market value of the asset to the replacement value of the asset [73].

We include control variables for suppliers' resources and capabilities that affect a firm's financial performance [74,75], including suppliers' size ($SIZE^S$), leverage (LEV^S), and sales growth rate ($SALEGR^S$), which are measured in the same way as the focal firm's size, leverage and sales growth rate. At the firm–supplier dyad level, we introduce the tenure of the relationship between a firm and its supplier ($TENURE^D$) into our model to control for the focal firm–supplier relationship strength following Song et al. [33].

The means, standard deviations, and correlations for all the variables are shown in Table 1. We test for multicollinearity via variance inflation factors (VIFs). The mean VIF is 1.88 and the maximum VIF is 4.32, both of which are below the recommended threshold of 5.0, which indicates that multicollinearity is not a concern in the following analysis. Additionally, we Winsorize all continuous variables at the 1 % level to

Table 1
Descriptive statistics and correlations.

Variables	Mean	SD	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
(1) *ROE*	0.038	0.062	1.000															
(2) *DIGIT*	0.078	0.161	0.088*	1.000														
(3) *PROXIMITY*	−1.359	1.024	0.097*	0.008	1.000													
(4) $SIZE^F$	22.889	2.250	0.185*	−0.106*	−0.060*	1.000												
(5) AGE^F	21.488	5.830	−0.052*	−0.032*	0.048*	0.071*	1.000											
(6) LEV^F	0.538	0.167	−0.461*	−0.174*	−0.062*	0.252*	0.175*	1.000										
(7) $CASH^F$	7.341	7.106	0.015*	−0.017*	−0.044*	0.135*	−0.035*	0.077*	1.000									
(8) $SALEGR^F$	0.104	0.263	0.329*	0.058*	0.073*	−0.024	−0.064*	−0.098*	0.068*	1.000								
(9) SRV^F	2.340	2.759	0.189*	0.127*	0.071*	−0.328*	−0.157*	−0.443	−0.290*	0.133*	1.000							
(10) PAY^F	5.430	4.874	0.111*	0.020	−0.019	0.037*	−0.067*	−0.137*	0.528*	0.108*	−0.147*	1.000						
(11) SHR^F	0.069	0.097	−0.070*	0.037*	0.009	−0.015	−0.059*	−0.014	0.042*	−0.068*	−0.068*	0.109*	1.000					
(12) TQ^F	1.665	0.969	0.391*	0.101*	0.076*	−0.146*	−0.112*	−0.308*	−0.112*	0.189*	0.713*	−0.031*	−0.072*	1.000				
(13) $SIZE^S$	7.471	2.055	−0.026	−0.044*	−0.024	0.047*	0.036*	0.074*	0.009	−0.005	−0.069*	0.022	0.001	−0.054*	1.000			
(14) LEV^S	0.413	0.203	−0.024	−0.048*	−0.003	0.020	0.066*	0.079*	0.023	0.008	−0.060*	0.013	0.001	−0.049*	0.602*	1.000		
(15) $SALEGR^S$	0.154	0.321	0.060*	−0.008	0.046*	0.019	−0.020	−0.018	−0.006	0.183	0.061*	−0.003	−0.016	0.058*	0.116*	0.081*	1.000	
(16) $TENURE^D$	2.366	1.259	−0.025	−0.076*	−0.302*	0.031*	−0.026	0.036*	0.069*	−0.015	−0.067*	0.049*	−0.014	−0.036*	0.084*	0.066*	−0.018	1.000

Note:
* $p < .05$.

address outliers.

3.3. Model

The following model was developed to test *H1*:

$$ROE_{i,t} = \beta_0 + \beta_1 DIGIT_{i,t} + \beta_2 DIGIT_{i,t}^2 + \beta_3 SIZE_{i,t}^F + \beta_4 AGE_{i,t}^F + \beta_5 LEV_{i,t}^F + \beta_6 CASH_{i,t}^F + \beta_7 SALEGR_{i,t}^F + \beta_8 SRV_{i,t}^F + \beta_9 PAY_{i,t}^F + \beta_{10} SHR_{i,t}^F + \beta_{11} TQ_{i,t}^F + \beta_{12} SIZE_{j,t}^S + \beta_{13} LEVERAGE_{j,t}^S + \beta_{14} SALEGR_{j,t}^S + \beta_{15} TENURE_{i,j,t}^D + u_i + v_t + \varepsilon_{i,j,t} \quad (3)$$

where $ROE_{i,t}$ denotes the financial performance of firm i in year t and $DIGIT_{i,t}$ denotes the digitalization level of firm i in year t. u and v are used to control for the firm and year fixed effects, respectively. Controlling for these two fixed effects can eliminate the influences of time-invariant firm characteristics and common shocks and trends at the time level, reducing omitted variable bias and effectively mitigating endogeneity issues. Finally, β_0 is a constant, and $\varepsilon_{i,j,t}$ denotes the error, which is clustered at the dyad level.

Furthermore, to test *H2*, the interaction terms among digitalization, the quadratic terms of digitalization and digitalization proximity are incorporated into Eq. (3) to obtain

$$ROE_{i,t} = \beta_0 + \beta_1 DIGIT_{i,t} + \beta_2 DIGIT_{i,t}^2 + \beta_3 PROXIMITY_{i,j,t} + \beta_4 DIGIT_{i,t} \times PROXIMITY_{i,j,t} + \beta_5 DIGIT_{i,t}^2 \times PROXIMITY_{i,j,t} + \beta_6 SIZE_{i,t}^F + \beta_7 AGE_{i,t}^F + \beta_8 LEV_{i,t}^F + \beta_9 CASH_{i,t}^F + \beta_{10} SALEGR_{i,t}^F + \beta_{11} SRV_{i,t}^F + \beta_{12} PAY_{i,t}^F + \beta_{13} SHR_{i,t}^F + \beta_{14} TQ_{i,t}^F + \beta_{15} SIZE_{j,t}^S + \beta_{16} LEVERAGE_{j,t}^S + \beta_{17} SALEGR_{j,t}^S + \beta_{18} TENURE_{i,j,t}^D + u_i + v_t + \varepsilon_{i,j,t} \quad (4)$$

In Eq. (4), the interaction terms capture how digitalization proximity moderates the nonlinear effect of firm digitalization on financial performance.

4. Analyses and results

4.1. Hypothesis testing

The findings of the hypothesis testing are summarized in Table 2. Specifically, Model 1 contains only the control variables, Model 2 introduces independent variables, and Model 3 further incorporates the moderator and interaction terms. The results show that most of the control variables affect financial performance significantly. In Model 2, the coefficient of the digitalization quadratic term is significantly negative ($\beta_2 = -0.0568$, $p < 0.01$) and that of digitalization is significantly positive ($\beta_1 = 0.0432$, $p < 0.01$). Furthermore, the lower- and upper-bound slopes are –0.0370 ($p < 0.05$) and 0.0258 ($p < 0.05$), respectively, and the inflection point (0.6574) is in the domain of the digitalization value, indicating an inverted U-shaped relationship between digitalization and financial performance [66], thereby supporting *H1*. Model 3 indicates the positive moderating effect of digitalization proximity. The significantly negative coefficient of the quadratic interaction term between digitalization and digitalization proximity ($\beta_5 = -0.0429$, $p < 0.05$) shows a steepening in the nonlinear curve [66]. Moreover, the inflexion point moves to the right as the digitalization proximity increases ($\beta_2\beta_5 - \beta_1\beta_4 = 0.0028 > 0$); specifically, high digitalization proximity complements the performance-enhancing effects of digitalization, enabling firms to achieve optimal performance outcomes more quickly. The above results provide empirical support for *H2*. To facilitate the visualization of digitalization proximity's moderating effect, we present the graphical representation in Fig. 2.

Table 2
Results of hypothesis testing.

ROE	(1)	(2)	(3)
DIGIT		0.0432*** (3.41)	0.0681*** (3.92)
$DIGIT^2$		–0.0568***	–0.1018**
		(–3.18)	(–3.83)
PROXIMITY			0.0001 (0.14)
DIGIT × *PROXIMITY*			0.0229** (2.21)
$DIGIT^2$ × *PROXIMITY*			–0.0429**
			(–2.22)
Firm–level controls			
$SIZE^F$	0.0017*** (4.26)	0.0016*** (4.10)	0.0016*** (4.02)
AGE^F	–0.0047 (–1.03)	–0.0054 (–1.18)	–0.0056 (–1.23)
LEV^F	–0.2641***	–0.2654***	–0.2645***
	(–29.85)	(–30.00)	(–29.87)
$CASH^F$	0.0009*** (5.76)	0.0010*** (5.80)	0.0010*** (5.84)
$SALEGR^F$	0.0589***	0.0592***	0.0590***
	(24.05)	(24.18)	(24.07)
SRV^F	–0.0020***	–0.0022***	–0.0021***
	(–3.35)	(–3.67)	(–3.57)
PAY^F	0.0008** (1.99)	0.0007* (1.87)	0.0007* (1.90)
SHR^F	–0.0328***	–0.0328***	–0.0328***
	(–3.33)	(–3.34)	(–3.34)
TQ^F	0.0093*** (6.28)	0.0096*** (6.49)	0.0096*** (6.43)
Supplier-level controls			
$SIZE^S$	0.0006** (2.32)	0.0006** (2.32)	0.0007** (2.36)
LEV^S	–0.0064**	–0.0062**	–0.0062**
	(–2.24)	(–2.17)	(–2.19)
$SALEGR^S$	0.0034** (2.38)	0.0034** (2.40)	0.0033** (2.36)
Dyad-level control			
$TENURE^D$	0.0006 (1.53)	0.0006 (1.47)	0.0007* (1.79)
Year fixed effect	Yes	Yes	Yes
Firm fixed effect	Yes	Yes	Yes
Constant	0.2149** (2.17)	0.2305** (2.33)	0.2354** (2.38)
Observations	4472	4472	4472
Within R^2	0.3245	0.3266	0.3276

Note: t statistics are shown in parentheses; standard error is clustered at the dyad level.
* $p < 0.1$,
** $p < 0.05$,
*** $p < 0.01$.

4.2. Robustness checks

We conduct a series of robustness checks to address potential endogeneity issues. First, reverse causality can induce endogeneity, thereby influencing the estimation outcomes. Although the theoretical evidence is limited, we recognize that profitable firms may be inclined to increase their investment in digitalization. Therefore, we employ a lagged model to resolve this problem. We lag our dependent variable by one year and re-estimate the models. The results provided in Table 3 are consistent with the main findings.

We also use the instrumental variable (IV) method with two-stage least squares (2SLS) estimators, which are frequently utilized in the literature. This method can address reverse causality and mitigate the interference of the second significant endogeneity problem, namely omitted variables. Specifically, following Larcker and Rusticus [76], we choose the average digitalization level of firms in the same year, city, and industry (excluding the designated focal firm; *AVED*) as the IV of digitalization. *AVED* fulfils two criteria for being an IV: correlation and exogeneity. That is, a firm's digitalization is correlated significantly with the average digitalization level of firms in a particular year and the city and industry in which it is located. This is because digitalization is influenced by technological developments over time. Institutional pressure also drives firms in the same industry and city to mimic each other. However, the average digitalization level in the same city, industry, and year does not affect the firm's financial performance directly because *AVED* is related to the institutional, technological, and business environments of a city and industry in a year; however, financial

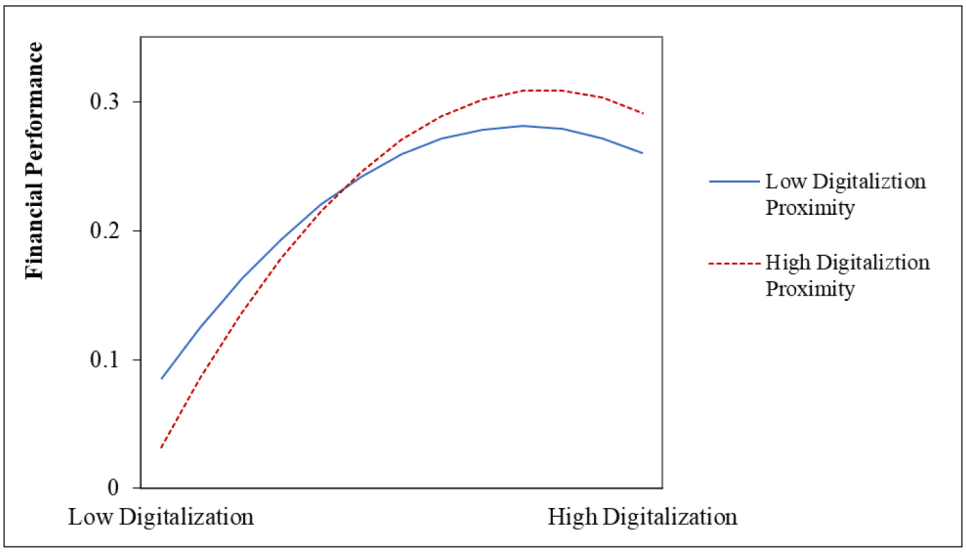


Fig. 2. The moderating effect of digitalization proximity.

Table 3
Results of the lagged model.

ROE_{t+1}	(1)	(2)
DIGIT	0.0758*** (3.75)	0.0914*** (4.38)
$DIGIT^2$	–0.0592** (–2.07)	–0.0785** (–2.64)
PROXIMITY		–0.0007 (–0.94)
DIGIT × *PROXIMITY*		0.0286*** (2.81)
$DIGIT^2$ × *PROXIMITY*		–0.0359** (–2.05)
Firm-level controls	Yes	Yes
Supplier-level controls	Yes	Yes
Dyad-level control	Yes	Yes
Year fixed effect	Yes	Yes
Firm fixed effect	Yes	Yes
Constant	0.5320*** (4.10)	0.5152*** (3.97)
Observations	2965	2965
Within R^2	0.1582	0.1613

Note: t statistics are shown in parentheses; standard error is clustered at the dyad level.
*$p < 0.1$,
**$p < 0.05$,
***$p < 0.01$.

performance is determined by a firm's resources, capabilities, operations, and strategies. Specifically, *AVED* is calculated via the equation

$$AVED_{r,k,t} = \frac{\Sigma_{p=1}^{n} DIGIT_{p,t} - DIGIT_{i,t}}{n-1}, i, p \in (r,k), \quad (5)$$

where *r* denotes the city, *t* is the year, and *k* is the industry. Then *p* represents the peer firms of focal firm *i* in the same year, industry, and city, and *n* is the total number of firms in the sample for the same year, industry, and city. As shown in Table 4, the results are consistent with those for the main models. Moreover, the IV passes the under-identification test and the weak identification test.

The third crucial endogeneity issue is sample selection bias. We employ Heckman's two-step method to address this issue, in line with Barker et al. [75]. In the first-step regression, we use $ISDIG_{i,b}$ a dummy variable, to capture the probability that a firm will invest in digitalization. We identify the digitalized intangible assets or construction in progress based on the keywords specified in Appendix 1 from the intangible assets and construction-in-progress items in the firm's annual financial reports. We set $ISDIG_{i,t}$ as 0 if firm *i* has no digitalized intangible assets or construction in progress in period *t* and 1 if it has. Next, we estimate a random effects probit model, where $ISDIG_{i,t}$ is regressed on the control variables that reflect a firm's resources (i.e., firm size,

Table 4
Results of 2SLS regression.

	First Stage		Second Stage	
	(1) *DIGIT*	(2) $DIGIT^2$	(3) *ROE*	(4) *ROE*
AVED	0.4634*** (13.06)	–0.0776*** (–3.17)		
$AVED^2$	0.4877*** (8.32)	1.0111*** (24.98)		
DIGIT			0.0646*** (2.66)	0.1151*** (2.70)
$DIGIT^2$			–0.0847** (–2.54)	–0.1721** (–2.57)
PROXIMITY				–0.0005 (–0.78)
DIGIT × *PROXIMITY*				0.0469** (2.38)
$DIGIT^2$ × *PROXIMITY*				–0.0831** (–2.07)
Firm-level controls	Yes	Yes	Yes	Yes
Supplier-level controls	Yes	Yes	Yes	Yes
Dyad-level control	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes
Firm fixed effect	Yes	Yes	Yes	Yes
Observations	4444	4444	4444	4444
LM statistics			1167.957 (0.00)	649.705 (0.00)
Wald F statistics			690.568	331.420
Centered R^2			0.3060	0.3065

Note: Standard error is clustered at the dyad level; *t*-values are shown in parentheses except for LM statistics, which have *p*-values in parentheses.
*$p < 0.1$,
**$p < 0.05$,
***$p < 0.01$.

firm age, cash turnover ratio, and sales growth rate), as a firm that has more resources tends to invest in digitalization. We also include the research and development (R&D) expenditure ratio ($RDER^F$), which is the ratio of R&D expenditures to total sales, as firms' willingness to invest in R&D. After the first-step regression, we generate a non-selection-hazard estimate and incorporate it into the second-step regression as an inverse Mills ratio (*IMR*; [77]). The results are

Table 5
Results of Heckman's two-step regression.

	First Stage	Second Stage	
	(1) *ISDIG*	(2) *ROE*	(3) *ROE*
$SIZE^F$	0.1305***		
	(11.81)		
AGE^F	–0.0072 (–1.53)		
$CASH^F$	–0.0141***		
	(–4.11)		
$SALEGR^F$	0.2583** (2.43)		
$RDER^F$	0.0167** (2.39)		
IMR		0.1367*** (5.80)	0.1357*** (5.76)
DIGIT		0.0414*** (3.26)	0.0650*** (3.73)
$DIGIT^2$		–0.0572***	–0.1002***
		(–3.19)	(–3.75)
PROXIMITY			0.0001 (0.20)
DIGIT × *PROXIMITY*			0.0219** (2.09)
$DIGIT^2$ × *PROXIMITY*			–0.0414**
			(–2.11)
Firm-level controls		Yes	Yes
Supplier-level controls		Yes	Yes
Dyad-level control		Yes	Yes
Year fixed effect	Yes	Yes	Yes
Firm fixed effect	Yes	Yes	Yes
Constant	–1.8487***	0.0486 (0.47)	0.0549 (0.53)
	(–4.16)		
Observations	4682	4436	4436
Pseudo/Within R^2	0.0673	0.3333	0.3343

Note: t statistics are shown in parentheses; standard error is clustered at the dyad level.
* $p < 0.1$,
** $p < 0.05$,
*** $p < 0.01$.

presented in Table 5 and are consistent with those of our main analysis.

Finally, there may be measurement errors in the calculation of main variables. To address concerns that the incomplete inclusion of investments in digitalization might distort the estimation results, we set a higher base threshold for the digitalization indicator [75]. Specifically, we re-estimate the model using a subsample with a 25 % base threshold applied. That is, samples with a value of digitalization less than 0.0015 were excluded, resulting in a reduced subsample of 3,316 observations. In addition, we assess our measurement reliability by using alternative measures that have been used in the previous literature [75,33]. Specifically, Chinese listed companies typically discuss and disclose their business strategies and development prospects in the management's discussion and analysis (MD&A) report. We thus compute word frequency related to digitalization from the MD&A report [78] and recalculate an alternative measure of digitalization (*MDAD*) and digitalization proximity (*MDAP*) via similar methods. For the dependent variable, we deploy a firm's return on assets (*ROA*), measured by the ratio of net profit to total assets, as an alternative measure for financial performance. The COVID-19 pandemic has drastically disrupted firms' operations, supply chains, and financial performance [60]. Therefore, we exclude samples from 2020 onward to avoid the pandemic's effect. These results are presented in Table 6 and are consistent with the previous findings.

4.3. Post hoc analyses

4.3.1. Firm–supplier digitalization asymmetry

Power reflects a firm's ability to influence its partner's actions and decisions in a supply chain [79]. It has been viewed as a key form of relational capital that influences supply chain behavior and performance [80]. A firm may use non-mediated or mediated power in supply chain relationships. Thus, we further explore how the power structure in the supply chain influences the relationship between digitalization and financial performance.

Table 6
Results of robustness tests to address measurement errors.

	25 % digitalization threshold		Alternative IV		Alternative DV		Alternative sample period	
	(1) *ROE*	(2) *ROE*	(3) *ROE*	(4) *ROE*	(5) *ROA*	(6) *ROA*	(7) *ROE*	(8) *ROE*
DIGIT	0.0300**	0.0586***			0.0459***	0.0722***	0.0477**	0.1087***
	(2.27)	(3.36)			(3.31)	(3.79)	(2.13)	(3.38)
$DIGIT^2$	–0.0360**	–0.0897***			–0.0559***	–0.1016***	–0.0753**	–0.2062***
	(–2.03)	(–3.58)			(–2.85)	(–3.49)	(–2.20)	(–3.79)
PROXIMITY		0.0001 (0.22)				0.0004 (0.63)		–0.0009 (–0.93)
DIGIT × *PROXIMITY*		0.0274***				0.0233** (2.06)		0.0570***
		(2.83)						(2.93)
$DIGIT^2$ × *PROXIMITY*		–0.0529***				–0.0424**		–0.1210***
		(–3.01)				(–2.00)		(–3.07)
MDAD			0.0029** (2.04)	–0.0003				
				(–0.17)				
$MDAD^2$			–0.0022***	–0.0012**				
			(–5.51)	(–2.18)				
MDAP				–0.0012				
				(–1.55)				
MDAD × *MDAP*				0.0021**				
				(2.13)				
$MDAD^2$ × *MDAP*				–0.0006**				
				(–2.44)				
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	0.2640**	0.2763** (2.16)	0.1140 (1.28)	0.1163 (1.31)	0.3297***	0.3381***	0.0689 (0.23)	0.1068 (0.35)
	(2.06)				(3.04)	(3.12)		
N	3316	3316	4596	4596	4472	4472	1755	1755
Within^2	0.2761	0.2790	0.3149	0.3162	0.3015	0.3029	0.1802	0.1856

Note: t statistics are shown in parentheses; standard error is clustered at the dyad level.
* $p < 0.1$,
** $p < 0.05$,
*** $p < 0.01$.

When a firm has the knowledge, expertise, or skills desired by its supply chain partner, it gives the firm non-mediated power over the partner [79]. Digitalization asymmetry reflects the difference in digital capabilities between a firm and its supplier. Thus, it influences the power structure in the supply chain and provides non-mediated power to parties with greater digital knowledge and capabilities [29,80]. We divide the sample into two groups according to the difference in the digitalization level between firms and their suppliers. We introduce a binary variable, *DA*, that takes a value of 1 when the digitalization level of a firm exceeds that of the corresponding supplier and 0 otherwise. Thus, the *DA* reflects the power structure in the supply chain. Specifically, $DA = 1$ indicates that the focal firm has non-mediated power over its supplier, and $DA = 0$ indicates that the supplier has non-mediated power over the focal firm.

The results of the analysis are reported in Table 7. When a firm leads digitalization in the supply chain (DA = 1), an inverted U-shaped relationship between digitalization and financial performance ($\beta_2 = -0.0525$, $p < 0.01$; $\beta_1 = 0.0456$, $p < 0.01$) and a positive moderating effect of digitalization proximity ($\beta_5 = -0.0416$, $p < 0.05$) still exist, which is consistent with the main analyses. We find that digitalization has a linear and positive effect on financial performance ($\beta_1 = 0.0753$, $p < 0.05$) and that an inverted U-shaped relationship does not exist ($\beta_2 = -0.1455$, $p > 0.1$) when the supplier is leading digitalization (DA = 0). The results also show that digitalization proximity enhances the impact of digitalization on financial performance ($\beta_4 = 0.0999$, $p < 0.01$).

4.3.2. Supplier concentration

When a firm can exert positive or negative reinforcements on its supply chain partners, it has mediated power over the partners [79]. Supplier concentration is measured by the proportion of a firm's spending on transactions with its top five suppliers relative to its cost of goods sold. A higher supplier concentration indicates that the major suppliers can increase or reduce the volume of business with the firm. Hence, it reflects the extent to which a firm depends on its suppliers [74,75]. Thus, supplier concentration influences the power structure in the supply chain, and high supplier concentration provides mediated power to major suppliers [80]. We divide the sample into two groups based on supplier concentration. The group with a supplier concentration above the mean value is designated the high-concentration group (*High*), in which major suppliers have mediated power. In contrast, the group with a supplier concentration below the mean value is designated the low-concentration group (*Low*), in which major suppliers do not have mediated power.

The results of the analysis are reported in Table 8. It is found that when supplier concentration is low, an inverted U-shaped relationship between digitalization and financial performance ($\beta_2 = -0.0943$, $p < 0.01$; $\beta_1 = 0.0725$, $p < 0.01$) and a positive moderating effect of digitalization proximity ($\beta_5 = -0.0686$, $p < 0.05$) still exist, which is consistent with the main analyses. When supplier concentration is high,

Table 8
Additional analysis for supplier concentration.

ROE	***High***		***Low***	
	(1)	**(2)**	**(3)**	**(4)**
DIGIT	–0.0281*	–0.0282	0.0725***	0.1176***
	(–1.79)	(–1.29)	(3.54)	(4.06)
$DIGIT^2$	0.0750***	0.0870***	–0.0943***	–0.1700***
	(3.39)	(2.43)	(–3.06)	(–3.81)
PROXIMITY		0.0001		0.0002 (0.27)
		(0.17)		
DIGIT × *PROXIMITY*		–0.0022		0.0387**
		(–0.18)		(2.28)
$DIGIT^2$ × *PROXIMITY*		0.0132		–0.0686**
		(0.51)		(–2.29)
Firm-level controls	Yes	Yes	Yes	Yes
Supplier-level controls	Yes	Yes	Yes	Yes
Dyad-level control	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes
Firm fixed effect	Yes	Yes	Yes	Yes
Constant	–0.0065	0.0025	0.2602**	0.2677**
	(–0.03)	(0.01)	(2.09)	(2.15)
Observations	2192	2192	2179	2179
Within R^2	0.3186	0.3189	0.2616	0.2643

Note: t statistics are shown in parentheses; standard error is clustered at the dyad level.
* $p < 0.1$,
** $p < 0.05$,
*** $p < 0.01$.

Table 7
Additional analysis for digitalization asymmetry.

ROE	***DA* = 1**		***DA* = 0**		
	(1)	**(2)**	**(3)**	**(4)**	**(5)**
DIGIT	0.0456***	0.0751***	0.0753**	0.0444	0.1602***
	(3.33)	(4.13)	(2.19)	(0.92)	(2.83)
$DIGIT^2$	–0.0525***	–0.0934***		–0.1455	0.1446
	(–2.91)	(–3.75)		(0.91)	(1.28)
PROXIMITY		–0.0006			–0.0005
		(–0.66)			(–1.15)
DIGIT × *PROXIMITY*		0.0271**			0.0999***
		(2.49)			(3.95)
$DIGIT^2$ × *PROXIMITY*		–0.0416**			
		(–2.17)			
Firm-level controls	Yes	Yes	Yes	Yes	Yes
Supplier-level controls	Yes	Yes	Yes	Yes	Yes
Dyad-level control	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes
Firm fixed effect	Yes	Yes	Yes	Yes	Yes
Constant	0.4159***	0.4160***	–1.8508***	–1.8533***	–1.8885***
	(3.85)	(3.85)	(–7.66)	(–7.67)	(–7.85)
Observations	2713	2713	1568	1568	1568
Within R^2	0.3236	0.3257	0.4258	0.4262	0.4329

Note: t statistics are shown in parentheses; standard error is clustered at the dyad level.
* $p < 0.1$,
** $p < 0.05$,
*** $p < 0.01$.

there is a U-shaped relationship between digitalization and financial performance ($\beta_2 = 0.0750$, $p < 0.01$; $\beta_1 = -0.0281$, $p < 0.1$), and the moderating effect of digitalization proximity is nonsignificant ($\beta_5 = 0.0132$, $p > 0.1$).

4.3.3. Mechanism test

We conduct a mechanism test to explore how digitalization affects firms' financial performance [49]. We argue that there is an inverted U-shaped relationship between digitalization and operational efficiency and that operational efficiency affects financial performance positively and linearly. Operational efficiency (*OPEE*) is measured by the ratio of operating revenue to total cost [59].

The results of the mechanism test are shown in Table 9. We find an inverted U-shaped relationship between digitalization and operational efficiency ($\beta_2 = -0.9754$, $p < 0.05$; $\beta_1 = 1.1929$, $p < 0.001$). The results reported in Model 2 ($\beta_3 = 0.0055$, $p < 0.001$) and Model 3 ($\beta_3 = 0.0054$, $p < 0.001$) suggest that there is a positive relationship between operational efficiency and financial performance. The impact of digitalization on financial performance is reduced when operational efficiency is considered. Hence, the findings reveal that digitalization influences financial performance through operational efficiency.

4.3.4. The impact of individual digital technology

To strengthen the validity of the findings and provide more nuanced insights into the impact of digitalization on financial performance, we decompose the measure of digitalization and develop measures for individual digital technologies (AI, blockchain, cloud computing, big data analytics, and digital applications) using a similar method. We examine their impacts on financial performance, and the results are reported in Table 10. It is found that with the exception of blockchain and big data analytics, the results for other technologies are consistent with the main analysis. That is, investments in AI ($\beta_1 = 0.0547, p < 0.1$; $\beta_2 = -0.0517, p < 0.1$), cloud computing ($\beta_1 = 0.5037, p < 0.01$; $\beta_2 = -0.0649, p < 0.05$), and digital applications ($\beta_2 = 0.0359$, $p < 0.01$; $\beta_2 = -0.0348$, $p < 0.1$) have an inverted U-shaped effect on firms' financial performance.

The relationship between investment in blockchain and financial performance is found to be significantly positive ($\beta_1 = 0.9992, p < 0.01$), with the quadratic term also showing a positive and significant effect ($\beta_2 = 0.2112, p < 0.01$). The inflection point of this curve ($x = -0.1057$) lies outside the sample data range, indicating that the relationship does not follow a U-shaped pattern but rather demonstrates an accelerating positive trend [66]. A possible explanation is that blockchain technology is still in its early stages of adoption, and its potential benefits are far from being fully realized [12]. As firms increase their investment in blockchain, they are likely to unlock new efficiencies, transparency, and trust-building mechanisms that compound over time, leading to disproportionately higher financial returns [81]. The fact that the mean value of investment in blockchain in the sample is the lowest for all digital technologies can also serve as important evidence (see Appendix 2). At current investment levels, most firms are likely operating on the upward-sloping portion of the blockchain value curve, where marginal returns are increasing rather than diminishing. This contrasts with more mature technologies, where over-investment and diminishing returns may already be evident.

In addition, the relationship between big data analytics and financial performance is statistically nonsignificant. This finding suggests that big data analytics alone may not be sufficient to drive significant financial value for firms in the current business landscape. A potential explanation is that big data analytics, while powerful, is not a standalone solution. Its effectiveness heavily depends on integrating complementary organizational resources, such as skilled personnel, advanced digital infrastructure, and a data-driven organizational culture [82,83]. As a data processing technology, big data analytics often needs to be combined with other digital technologies that can collect high-quality data to unlock its full potential. Furthermore, the growing adoption of big data platforms such as Amazon Web Services (AWS) has led firms to increasingly outsource their analytics capabilities. This reliance on external platforms reduces the performance advantage traditionally associated with in-house big data analytics.

Table 9
The mechanism test.

	(1) *OPEE*	(2) *ROE*	(3) *ROE*
DIGIT	1.1926*** (3.36)		0.0360*** (2.95)
$DIGIT^2$	–0.9800** (–1.98)		–0.0385** (–2.26)
OPEE		0.0055*** (10.16)	0.0054*** (9.93)
Firm-level controls	Yes	Yes	Yes
Supplier-level controls	Yes	Yes	Yes
Dyad-level control	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes
Firm fixed effect	Yes	Yes	Yes
Constant	–8.8456*** (–3.08)	0.1383 (1.47)	0.1548** (1.64)
Observations	4598	4598	4598
Within R^2	0.0173	0.3260	0.3278

Note: t statistics are shown in parentheses; standard error is clustered at the dyad level.
* $p < 0.1$,
** $p < 0.05$,
*** $p < 0.01$.

Table 10
The impact of individual digital technology.

ROE	(1)	(2)	(3)	(4)	(5)
AI	0.0547* (1.85)				
AI^2	–0.0517* (–1.71)				
Blockchain		0.9992*** (4.32)			
$Blockchain^2$		0.2112*** (3.94)			
Cloudcomp			0.5037*** (2.76)		
$Cloudcomp^2$			–0.0649** (–2.00)		
Bigdata				0.0194 (0.52)	
$Bigdata^2$				0.0114 (0.42)	
Digitalapp					0.0359*** (2.64)
$Digitalapp^2$					–0.0348* (1.87)
Controls	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Constant	0.0919 (0.96)	0.0848 (0.89)	0.0672 (0.70)	0.0920 (0.96)	0.1005 (1.06)
Observations	4598	4598	4598	4598	4598
Within R^2	0.3094	0.3148	0.3102	0.3089	0.3103

Note: t statistics are shown in parentheses; standard error is clustered at the dyad level.
* $p < 0.1$,
** $p < 0.05$,
*** $p < 0.01$.

5. Discussion and conclusions

5.1. Discussion of findings

The empirical analyses document how a firm's digitalization affects its financial performance and the impact of digitalization proximity on

the relationship. First, the results clarify the complex relationship between digitalization and financial performance [23,27]. We find an inverted U-shaped relationship, which reveals that moderate levels of digitalization can lead to the best financial performance. This finding coincides with digitalization's "too-much-of-a-good-thing" argument [23,84]. It also provides a possible explanation for the digitalization paradox phenomenon [49,52,53].

Second, the results suggest that suppliers play critical roles in helping firms benefit from digitalization. Increasing digitalization proximity between a firm and its supplier steepens the inverted U-shaped curve between the firm's digitalization and financial performance and shifts the inflexion point to the right. The findings reveal that proximity in digital technology investment between a firm and its supplier plays a complementary role in leveraging the performance-enhancing effects of digitalization. These findings corroborate the argument that strategic fit with supply chain partners in the digitalization investment is essential for benefiting from digital technologies [85].

Third, the post hoc analyses reveal that when a firm has greater digital capabilities than its suppliers or when supplier concentration is low, there is still an inverted U-shaped relationship between digitalization and financial performance, which is moderated positively by digitalization proximity. A firm with a higher digitalization level has non-mediated power, which is relational and associated with a positive attitude toward supply chain relationships [79]. Its influence is exerted indirectly and is less manipulative. Suppliers can decide whether and how much to be influenced by non-mediated power. Moreover, when supplier concentration is low, major suppliers do not have the mediated power to influence firms' operations and decisions. Thus, the power structure in the supply chain does not influence the main findings in these two situations.

When a supplier has greater digital capabilities than a firm, the supplier has non-mediated power. The extent to which a firm is influenced by a supplier's use of non-mediated power is determined by the firm [80]. Thus, a firm can learn from the supplier's expertise and skills and invest in digitalization when it benefits its operational and supply chain processes. A firm's continued investment in digitalization can improve the efficiency and effectiveness of collaboration in supply chains, enhancing its financial performance. This finding indicates that a firm can benefit from a supplier's use of non-mediated power and learning from the supplier, as digitalization proximity strengthens the positive impact of digitalization on financial performance. Thus, collaborating with suppliers with greater digital capabilities could be one method of addressing the digitalization paradox.

Mediated power can be exerted according to the intention of the power holder [80]. When supplier concentration is high, major suppliers decide whether, when, and how to use mediated power to influence a firm's decisions on digitalization. Thus, major suppliers may use mediated power to push the firm to invest in digitalization even when its employees do not have the skills and knowledge or when it does not have a management system to support the implementation of the technologies. As a result, digitalization may reduce financial performance. The firm can accumulate digital knowledge and skills, hire employees with digital capabilities, and develop innovative processes during its digitalization journey, allowing it to benefit from digitalization. Hence, there is a U-shaped relationship between digitalization and financial performance. Mediated power is manipulative and instrumental. It is a competitive use of power. Thus, a firm tends to reduce its dependence on major suppliers by adopting digital technologies different from those used by major suppliers. Moreover, major suppliers may use mediated power to push a focal firm to adopt similar digital technologies. As a result, the moderating effect of digitalization proximity becomes insignificant.

5.2. Research implications

This study contributes to the literature by investigating the relationship between digitalization and financial performance. Research has reported positive effects of digitalization on performance outcomes [18–20,37]. However, some studies have acknowledged the potential risks of digitalization [21,22,26], such as the digitalization paradoxes [52,53]. Unexpected empirical evidence on digitalization investment has prompted scholars to examine the complex relationship between digitalization and performance [23–25,27]. Using a sample of Chinese listed companies, we contribute to this line of literature by providing empirical evidence of an inverted U-shaped relationship between digitalization and financial performance and revealing the mechanism through which digitalization affects firms' financial performance. We show that digitalization could be a double-edged sword and that mindlessly investing in digital technologies could influence a firm's competitive advantage negatively. Thus, we suggest that researchers consider nonlinear relationships when investigating the impact of digitalization on performance outcomes.

Our study also contributes to the literature by taking a supply chain perspective and linking digitalization proximity to suppliers with the performance outcomes of a firm's digitalization. We find that digitalization proximity is a critical factor influencing the inverted U-shaped relationship between digitalization and financial performance. This finding aligns with those of previous studies that focused on the positive role of technology proximity (e.g., [33]). Studies have focused on understanding how digitalization influences firm performance from an internal perspective [19,23,41]. This discovery highlights the significance of supply chains affecting the success of a firm's digitalization strategy [86,87] and reveals that the effectiveness of digitalization is contingent on alignment with suppliers' digitalization strategies [29,88]. Moreover, the results of the post hoc analyses enhance the understanding of the boundary conditions of the digitalization–financial performance relationship. The findings reveal that the power structure in supply chains influences the impact of digitalization on financial performance. We also find that the mediated and non-mediated power between a firm and its supplier play different roles. The results provide additional evidence that a supply chain perspective is essential for firms to benefit from digitalization. The findings offer insights into why digitalization paradoxes exist. We suggest that researchers consider supply chain factors such as digitalization proximity, firm–supplier digitalization asymmetry, and supplier concentration when investigating the impact of digitalization on financial performance empirically.

5.3. Managerial implications

The finding of an inverted U-shaped relationship between digitalization and financial performance offers valuable insights into Chinese firms' digitalization strategies. This highlights the importance of managing digitalization efforts strategically to maximize financial benefits while avoiding potential pitfalls. During the initial phases of digitalization, firms can improve their financial performance significantly. According to the analysis, approximately 90 % of the firms in the sample are below the inflection point. Thus, most Chinese firms can still benefit from digitalization. We suggest that firms invest in advanced digital technologies such as AI, blockchain, cloud computing, and digital applications. However, managers must recognize that digitalization efforts may reach a point of diminishing returns, where further investments can lead to diminishing financial returns or even reduced performance. We suggest that firms monitor their investment in digital technologies carefully (e.g., AI, cloud computing, and digital applications) and consider the tradeoffs between benefits and costs when developing digital strategies. The analysis reveals that the best digitalization level for Chinese firms is 0.6563, which indicates that 65.63 % of intangible assets and construction in progress are digitalized intangible assets and construction in progress. Thus, we suggest that Chinese firms increase investment in digital technologies if the ratio of digitalized intangible assets and construction in progress to total intangible assets and total construction in progress is lower than 0.6563 and reduce investment

otherwise.

We suggest that firms consider suppliers' investment in digital technologies. Managers must recognize that digitalization proximity to suppliers enables firms to realize the benefits of digitalization fully. Thus, we suggest that firms collaborate with suppliers to align digitalization strategies and investment in digital technologies. Additionally, we suggest that firms select suppliers that have adopted a similar digitalization strategy when making sourcing decisions. Firms should also design supplier development programs to help suppliers adopt and implement digital technologies aligned with firms. Additionally, we recommend that managers involve suppliers in discussions and management of digital strategies.

We suggest that Chinese firms consider the power structure within the supply chain when designing a digitalization strategy. Managers should be aware that digitalization asymmetry influences the digitalization–financial performance relationship. Firms should select and collaborate with suppliers that have high digitalization capabilities. Managers should be aware that collaborating with those suppliers could provide a solution to the digitalization paradox. We also recommend that firms increase their investment in digital technologies when they have a high level of supplier concentration. Managers should be aware that even if the initial investment in digital technologies may be negatively associated with financial performance, they should invest continuously in digitalization. They can benefit from it after they achieve a certain level of digitalization.

5.4. Limitations and future research

This paper has the following limitations that lead to future research opportunities. First, our findings are based on a sample of Chinese manufacturers. These findings may be influenced by China's business, institutional, and cultural environment. Thus, future research could generalize and validate the findings in other countries. Second, this study focuses on digitalization proximity to suppliers. Other supply chain stakeholders, such as customers, could also influence the impact of digitalization on financial performance. Future research could take a network perspective and explore how digitalization proximity at the network level influences a firm's digitalization. Third, the analysis explores only the impacts of the power structure on the relationship between digitalization and financial performance. Future studies could take a relational perspective and investigate how supply chain relationships, such as trust, influence the performance outcome of digitalization. Fourth, this study focuses on a supply chain perspective when investigating the contingencies of the impact of digitalization on financial performance. Future studies could investigate how the heterogeneity of firm characteristics, such as ownership, size and R&D intensity, influences the relationship between digitalization and performance outcomes. Fifth, this study focuses on the impact of a firm's digitalization on its financial performance. Future research could extend this study by investigating how a firm's digitalization affects supply chain performance such as resilience and sustainability.

Funding Source

This research is supported by the Fundamental Research Funds for the Central Universities, [UIBE (ZD3-11)].

CRediT authorship contribution statement

Yinan Qi: Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Kexin Hou:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Min Zhang:** Writing – review & editing, Validation, Resources, Project administration. **Tianlong Pu:** Writing – review & editing. **Hangfei Guo:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix 1. Selected studies on the impact of digitalization on performance

Study	Method	Measure for digitalization	Sample	Finding	Moderator
Abou-Foul et al. [46]	Questionnaire survey	A nine-item instrument encapsulated two dimensions: digital components embedded in physical products and digital capabilities embedded in the operational processes	185 U.S. and European manufacturing firms	Positive digitalization–financial performance relationship	—
Bhandari et al. [27]	OLS regression model	A digitalization score is calculated by annual word count divided by total words for every annual report and normalized to a scale from 0 to 1	571 US-listed manufacturing firms from 1992 to 2019	U-shaped digitalization–performance relationship	Net-FDI inflow (% of GDP)
Kohtamäki et al. [26]	Questionnaire survey	An instrument related to digital sales support, digital service support, digital analysis support, and data integration and access support	131 Swedish manufacturing firms	Inverted U-shaped digitalization–financial performance relationship	Servitization: consisting of operational services, R&D services, and consulting services
Li et al. [47]	Questionnaire survey	A four-item instrument captured how a firm uses digital technologies to understand customers, make decisions, increase added value, and launch new business models.	192 Chinese manufacturing firms	Positive digitalization–financial performance relationship	Knowledge inertia: A firm's proclivity to address difficulties based on prior experience and knowledge
Li et al. [48]	Questionnaire survey	External and internal digitalization: how a firm used digital technologies to improve social relations and process efficiency	181 Chinese firms	Positive internal/external digitalization–financial performance relationship	Informational diversity; shared vision
Yang & Yee [20]	Event study approach	Manufacturers' list of public announcements on the implementation of process digitalization initiative	168 Chinese listed firms over the past 13 years (i.e., 2006–2018)	Positive relationship between process digitalization, initiative, and profitability	Differential strategy; absorptive capacity; lean production

(*continued on next page*)

(*continued*)

Study	Method	Measure for digitalization	Sample	Finding	Moderator
Zeng et al. [49]	Two-way fixed effects model	Text mining method to extract keywords related to digitalization from the annual reports of firms	A-share listed companies in China, 16863 observations between 2012 and 2019	Positive digitalization–financial performance relationship	TMT experience; industry competition
Zomer et al. [45]	Structural equation modelling	The percentage of investments in digital concerning all of the investments made by the company	427 large global companies	Positive digitalization capability–business performance relationship	—
This study	Two-way fixed effects model	The ratio of the firm's digitalized intangible assets and digitalized construction in progress to total intangible assets and total construction in progress	4472 firm-supplier-year observations, including 788 Chinese listed manufacturing firms and 1448 suppliers from 2012 to 2023	Inverted U-shaped digitalization–financial performance relationship	Digitalization proximity between the focal firm and its suppliers

Appendix 2. Feature lexicon for digitalization

Category	Keywords	Mean	SD
Artificial Intelligence	artificial intelligence, business intelligence, image understanding, investment decision support system, intelligent data analysis, intelligent robot, machine learning, deep learning, semantic search, biometric technology, face recognition, speech recognition, authentication, autopilot, natural language processing	0.0073	0.0319
Big data analytics	big data, data mining, text mining, data visualization, heterogeneous data, credit investigation, augmented reality, mixed reality, virtual reality	0.0049	0.0369
Cloud Computing	cloud computing, stream computing, graph computing, in-memory computing, secure multi-party computation, brain-inspired computing, green computing, cognitive computing, converged architecture, billion-level concurrency, EB-level storage, Internet of things, cyber-physical systems	0.0007	0.0050
Blockchain	blockchain, digital currency, distributed computing, differential privacy, smart financial contracts	0.0005	0.0031
Digital Applications	mobile Internet, industrial Internet, Internet plus, mobile connectivity, digital healthcare, electronic commerce, mobile payment, third party payment, NFC payment, smart energy, smart wearable, smart agriculture, smart healthcare, smart home, smart cultural tourism, smart environmental protection, smart grid, smart marketing, B2B, B2C, C2B, C2C, O2O, intelligent transportation system, intelligent customer service, intelligent investment advisory, unmanned retail, digital finance, fintech, quantitative finance, open banking	0.0578	0.1432

Note: Mean and SD show the mean value and the standard deviation of the digitalization level in terms of artificial intelligence, big data analytics, cloud computing, blockchain, and digital applications, respectively.

References

[1] Y. Chen, M. Zhang, L. Mattews, H. Guo, Digital transformation and environmental information disclosure in China: the moderating role of top management team's ability, Bus. Strategy. Environ. (2024), https://doi.org/10.1002/bse.3930.
[2] M. Fitzgerald, N. Kruschwitz, D. Bonnet, M. Welch, Embracing digital technology: A new strategic imperative, MIT Sloan. Manage Rev. 55 (2) (2014) 1–12.
[3] M. Matarazzo, L. Penco, G. Profumo, R. Quaglia, Digital transformation and customer value creation in made in Italy SMEs: A dynamic capabilities perspective, J. Bus. Res. 123 (2021) 642–656.
[4] L.S. Dalenogare, B.G. Beniteza, N.F. Ayalab, A.G. Frank, The expected contribution of industry 4.0 technologies for industrial performance, Int. J. Prod. Econ. 204 (2018) 383–394.
[5] H. Gebauer, E. Fleisch, C. Lamprecht, F. Wortmann, Growth paths for overcoming the digitalization paradox, Bus. Horiz. 63 (2020) 313–323.
[6] B. Jovanovic, P.L. Rousseau, General purpose technologies, NBER Working Paper Series, working paper 11093. http://www.nber.org/papers/w11093, 2005.
[7] S. Mithas, Z.-L. Chen, T.J.V. Saldanha, A. De Oliveira Silveira, How will artificial intelligence and industry 4.0 emerging technologies transform operations management? Prod. Oper. Manage 31 (12) (2022) 4475–4487.
[8] M. Yang, M. Fu, Z. Zhang, The adoption of digital technologies in supply chains: drivers, process and impact, Technol. Forecast. Soc. Change 169 (2021) 120795.
[9] T. Choi, S. Kumar, X. Yue, H. Chan, Disruptive technologies and operations management in the industry 4.0 era and beyond, Prod. Oper. Manag. 31 (1) (2022) 9–31.
[10] A.G. Frank, L.S. Dalenogare, N.F. Ayala, Industry 4.0 technologies: implementation patterns in manufacturing companies, Int. J. Prod. Econ. 210 (2019) 15–26.
[11] L.X. Li, Y. Tong, L. Wei, S.L. Yang, Digital technology-enabled dynamic capabilities and their impacts on firm performance: evidence from the COVID-19 pandemic, Inf. Manage 59 (8) (2022) 103689.
[12] N. Upadhyay, Demystifying blockchain: A critical analysis of challenges, applications and opportunities, Int. J. Inf. Manag. 54 (2020) 102120.
[13] Z.S. Zhou, Z. Li, Corporate digital transformation and trade credit financing, J. Bus. Res. 160 (2023) 113793.
[14] T. Bresnahan, General purpose technologies, Handb. Econ. Innov. 2 (2010) 761–791.
[15] G. Vial, Understanding digital transformation: A review and a research agenda, J. Strateg. Inf. Syst. 28 (2) (2019) 118–144.
[16] X.C. Guo, M.M. Li, Y.L. Wang, A. Mardani, Does digital transformation improve the firm's performance? From the perspective of digitalization paradox and managerial myopia, J. Bus. Res. 163 (2023) 113868.
[17] Accenture, Digital Transformation: A sustainable evolution Journey, from. https://www.accenture.com/cn-en/insights/strategy/china-digital-transformation-index-2022, 2022.
[18] L. Ardito, S. Raby, V. Albino, B. Bertoldi, The duality of digital and environmental orientations in the context of SMEs: implications for innovation performance, J. Bus. Res. 123 (2021) 44–56.
[19] M. Tian, Y. Chen, G.H. Tian, W. Huang, C. Hu, The role of digital transformation practices in the operations improvement in manufacturing firms: A practice-based view, Int. J. Prod. Econ. 262 (2023) 108929.
[20] Y.F. Yang, R.W.Y. Yee, The effect of process digitalization initiative on firm performance: a dynamic capability development perspective, Int. J. Prod. Econ. 254 (2022) 108654.
[21] P. Smith, M. Beretta, The Gordian knot of practicing digital transformation: coping with emergent paradoxes in ambidextrous organizing structures, J. Prod. Innov. Manag. 38 (1) (2021) 166–191.
[22] H. Wimelius, L. Mathiassen, J. Holmström, M. Keil, A paradoxical perspective on technology renewal in digital transformation, Inf. Syst. J. 31 (1) (2021) 198–225.
[23] Y. Gong, Y. Yao, A. Zan, The too-much-of-a-good-thing effect of digitalization capability on radical innovation: The role of knowledge accumulation and knowledge integration capability, J. Knowl. Manag. 27 (6) (2023) 1680–1701.
[24] X.D. Xiang, L. Jiang, Digitalization and commercial bank performance: a test of heterogeneity from Chinese commercial banks, Finance Res. Lett. (2023) 104303.
[25] L.X. Li, Digital transformation and sustainable performance: the moderating role of market turbulence, Ind. Mark. Manag. 104 (2022) 28–37.
[26] M. Kohtamäki, V. Parida, P.C. Patel, H. Gebauer, The relationship between digitalization and servitization: The role of servitization in capturing the financial potential of digitalization, Technol. Forecast. Soc. Change 151 (2020) 119804.
[27] R.K. Bhandari, P. Zamborský, M. Ranta, J. Salo, Digitalization, internationalization, and firm performance: A resource-orchestration perspective on new OLI advantages, Int. Bus. Rev. (2023) 102135.
[28] J. Holmström, M. Holweg, B. Lawson, F.K. Pil, S.M. Wagner, The digitalization of operations and supply chain management: theoretical and methodological implications, J. Oper. Manag. 65 (2019) 728–734.

[29] B.G. Son, H. Kim, D. Hur, N. Subramanian, The dark side of supply chain digitalization: supplier-perceived digital capability asymmetry, buyer opportunism and governance, Int. J. Oper. Prod. Manag. 41 (7) (2021) 1220–1247.
[30] L.X. Li, X.S. Xu, Y. Ning, Y.q. Liu, S.L. Yang, How should companies deploy their digital supply chain platforms to gain competitive advantages? asset orch. perspect. Inf. Manag. 60 (6) (2023) 103842.
[31] J.B. Lin, S.Z. Lin, J. Benitez, X. Luo, A. Ajamieh, How to build supply chain resilience: the role of fit mechanisms between digitally-driven business capability and supply chain governance, Inf. Manage 60 (2) (2023) 103747.
[32] M. Benner, J. Waldfogel, Close to you? Bias and precision in patent-based measures of technological proximity, Res. Policy 37 (2008) 1556–1567.
[33] S. Song, Y. Dong, T. Kull, C. Carter, K. Xu, Supply chain leakage of greenhouse gas emissions and supplier innovation, Prod. Oper. Manage 32 (2023) 882–903.
[34] C. Wang, Q. Hu, Knowledge sharing in supply chain networks: effects of collaborative innovation activities and capability on innovation performance, Technovation 94 (2020) 102010.
[35] L.V. Lerman, G.B. Benitez, J.M. Müller, P.R. De Sousa, A.G. Frank, Smart green supply chain management: A configurational approach to enhance green performance through digital transformation, Supply Chain Manag.: Int. J. 27 (7) (2022) 147–176.
[36] C.L. Zhu, Y.H. Xiao, R.X. Wang, How smart shall we be? Optimal distinctiveness of digitalization and innovation performance, IEEE Trans. Eng. Manage 71 (2023) 8747–8760.
[37] M. Nasiri, M. Saunila, J. Ukko, Digital orientation, digital maturity, and digital intensity: determinants of financial success in digital transformation settings, Int. J. Oper. Prod. Manag. 42 (13) (2022) 274–298.
[38] A. Bharadwaj, O.A. ElSawy, P.A. Pavlou, N. Venkatraman, Digital business strategy: toward a next generation of insights, MIS Q. 37 (2) (2013) 471–482.
[39] C. Loebbecke, A. Picot, Reflections on societal and business model transformation arising from digitization and big data analytics, J. Strateg. Inf. Syst. 24 (2015) 149–157.
[40] S. Nambisan, K. Lyytinen, A. Majchrzak, M. Song, Digital innovation management: reinventing innovation management research in a digital world, MIS Q. 41 (1) (2017) 223–238.
[41] G.Q. Yang, Y.M. Nie, H.G. Li, H.S. Wang, Digital transformation and low-carbon technology innovation in manufacturing firms: the mediating role of dynamic capabilities, Int. J. Prod. Econ. 263 (2023) 108969.
[42] R. Dubey, A. Gunasekaran, S.J. Childe, T. Papadopoulos, Z.W. Luo, S.F. Wamba, D. Roubaud, Can big data and predictive analytics improve social and environmental sustainability? Technol. Forecast. Soc. Change 144 (2019) 534–545.
[43] C. Cai, Y.Q. Tu, Z. Li, Enterprise digital transformation and ESG performance, Finance Res. Lett. (2023) 104692.
[44] E. Brynjolfsson, L. Hitt, Information technology as a factor of production: the role of differences among firms, Econ. Innov. New Technol. 3 (3-4) (1995) 183–200.
[45] T. Zomer, A. Neely, V. Martinez, Digital transforming capability and performance: a micro-foundational perspective, Int. J. Oper. Prod. Manag. 40 (7-8) (2020) 1095–1128.
[46] M. Abou-Foul, J.L. Ruiz-Alba, A. Soares, The impact of digitalization and servitization on the financial performance of a firm: an empirical analysis, Prod. Plan. Control 32 (12) (2021) 975–989.
[47] L.X. Li, F. Ye, Y.Z. Zhan, A. Kumar, F. Schiavone, Y.N. Li, Unraveling the performance puzzle of digitalization: evidence from manufacturing firms, J. Bus. Res. 149 (2022) 54–64.
[48] L.X. Li, W. Tang, H.D. Zhou, S.L. Yang, Digitalization and firm performance: the moderating role of top management team attributes, IEEE Trans. Eng. Manage (2024) early access.
[49] H.X. Zeng, H.X. Ran, Q. Zhou, Y.L. Jin, X. Cheng, The financial effect of firm digitalization: evidence from China, Technol. Forecast. Soc. Change 183 (2022) 121951.
[50] S. Bar-Gill, E. Brynjolfsson, N. Hak, Helping small businesses become more data-driven: A field experiment on eBay, Manag. Sci. 70 (11) (2024) 7345–7372.
[51] E. Brynjolfsson, K. McElheran, The rapid adoption of data-driven decision-making, Am. Econ. Rev. 106 (5) (2016) 133–139.
[52] D. Sjödin, V. Parida, M. Kohtamäki, J. Wincent, An agile co-creation process for digital servitization: A micro-service innovation approach, J. Bus. Res. 112 (2020) 478–491.
[53] K.S.R. Warner, M. Wäger, Building dynamic capabilities for digital transformation: an ongoing process of strategic renewal, Long Range Plan. 52 (3) (2019) 326–349.
[54] E. Brynjolfsson, D. Rock, C. Syverson, The productivity J-curve: how intangibles complement general purpose technologies, Am. Econ. J.: Macroecon. 13 (1) (2021) 333–372.
[55] D.A. Vaio, R. Palladino, A. Pezzi, D.E. Kalisz, The role of digital innovation in knowledge management systems: a systematic literature review, J. Bus. Res. 123 (2021) 220–231.
[56] D. Ma, Q. Zhu, Innovation in emerging economies: research on the digital economy driving high-quality green development, J. Bus. Res. 145 (2022) 801–813.
[57] F.S. Wamba, A. Gunasekaran, S. Akter, S.J. Ren, R. Dubey, S.J. Childe, Big data analytics and firm performance: effects of dynamic capabilities, J. Bus. Res. 70 (2017) 356–365.
[58] S. Cao, H. Yao, M. Zhang, CSR gap and firm performance: an organizational justice perspective, J. Bus. Res. 158 (2023) 113692.
[59] L. Gao, L. Wan, Does corporate environmental responsibility contribute to financial performance? A dual path analysis through operational efficiency and the cost of debt, Corp. Soc. Responsib. Environ. Manag. 30 (1) (2022) 308–323.
[60] Y. Qi, X. Wang, M. Zhang, Q. Wang, Developing supply chain resilience through integration: an empirical study on an e-commerce platform, J. Oper. Manag. 69 (3) (2023) 477–496.
[61] F. Cappa, R. Oriani, E. Peruffo, Big data for creating and capturing value in the digitalized environment: unpacking the effects of volume, variety, and veracity on firm performance, J. Prod. Innov. Manag. 38 (1) (2021) 49–67.
[62] N. Li, Q. Yao, H.J. Tang, Y. Lu, Is digitalization necessary? Configuration of supply chain capabilities for improving enterprise competitive performance, J. Bus. Res. 186 (2025) 114972.
[63] F.M. Somohano-Rodriguez, A. Madrid-Guijarro, J.M. Lopez-Fernandez, Does Industry 4.0 really matter for SME innovation? J. Small Bus. Manag. 60 (4) (2020) 1001–1028.
[64] B. Kindermann, S. Beutel, G. de Lomana, S. Strese, D. Bendig, M. Brettel, Digital orientation: conceptualization and operationalization of a new strategic orientation, Eur. Manag. J. 39 (5) (2021) 645–657.
[65] R.S. Kaplan, D.P. Norton, Alignment: Using the balanced scorecard to create corporate synergies, Harvard Business Press, Boston, MA, 2006.
[66] R.F.J. Haans, C. Pieters, Z.-L. He, Thinking about U: theorizing and testing U- and inverted U-shaped relationships in strategy research, Strateg. Manage J. 37 (2016) 1177–1195.
[67] G. Culot, M. Podrecca, G. Nassimbeni, G. Orzes, M. Sartor, Using supply chain databases in academic research: A methodological critique, J. Supply Chain Manag. 59 (1) (2023) 3–25.
[68] F. Tao, X.R. Wang, Y. Xu, P. Zhu, Digital transformation, resilience of industrial chain and supply chain, and enterprise productivity, China Ind. Econ. 5 (2023) 118–136 (in Chinese).
[69] C.M. Li, P. Huo, Z.Y. Wang, W.G. Zhang, F.Y. Liang, A. Mardani, Digitalization generates equality? Enterprises' digital transformation, financing constraints, and labor share in China, J. Bus. Res. 163 (2023) 113924.
[70] N.V. Stein, N. Sick, J. Leker, How to measure technological distance in collaborations - the case of electric mobility, Technol. Forecast. Soc. Change 97 (2015) 154–167.
[71] S. Steinker, M. Pesch, K. Hoberg, Inventory management under financial distress: an empirical analysis, Int. J. Prod. Res. 54 (17-18) (2016) 5182–5207.
[72] M. Bagirov, C. Mateus, Oil prices, stock markets and firm performance: evidence from Europe, Int. Rev. Econ. Finance 61 (MAY) (2019) 270–288.
[73] M. Vizcaíno, J.P. Chousa, Analyzing the influence of the funds' support on Tobin's q using SEM and fsQCA, J. Bus. Res. 69 (2016) 2118–2124.
[74] Y. Chu, X. Tian, W. Wang, Corporate innovation along the supply chain, Manag. Sci. 65 (6) (2019) 2445–2466.
[75] J.M. Barker, C. Hofer, K. Hoberg, C. Eroglu, Supplier inventory leanness and financial performance, J. Oper. Manag. 68 (4) (2022) 385–407.
[76] D.F. Larcker, O.T. Rusticus, On the use of instrumental variables in accounting research, J. Account. Econ. 49 (3) (2010) 186–205.
[77] T.S. Certo, J.R. Busenbark, H.S. Woo, M. Semadeni, Sample selection bias and Heckman models in strategic management research, Strateg. Manage J. 37 (13) (2016) 2639–2657.
[78] C. Yuan, T.S. Xiao, C.X. Geng, Y. Sheng, Digital transformation and division of labor between enterprises: vertical specialization or vertical integration, China Ind. Econ. 9 (2021) 137–155 (In Chinese).
[79] X. Zhao, B. Huo, B.B. Flynn, J.H.Y. Yeung, The impact of power and relationship commitment on the integration of manufacturers and customers in a supply chain, J. Oper. Manag. 26 (3) (2008) 368–388.
[80] M. Zhang, X. Zhao, B. Huo, B. Flynn, The impact of power and relationship commitment on customer integration: a replication and extension, Int. J. Phys. Distrib. Logist. Manag. 56 (3) (2022) 238–260.
[81] P. Centobelli, R. Cerchione, P.D. Vecchio, E. Oropallo, G. Secundo, Blockchain technology for bridging trust, traceability and transparency in circular supply chain, Inf. Manage. 59 (2022) 103508.
[82] P. Mikalef, J. Krogstie, Examining the interplay between big data analytics and contextual factors in driving process innovation capabilities, Eur. J. Inf. Syst. 29 (3) (2020) 260–287.
[83] Y. Yeh, R. Eden, E. Fielt, R. Syed, The role of use for the business value of big data analytics, J. Strateg. Inf. Syst. 34 (2025) 101888.
[84] M.M. Feliciano-Cestero, N. Ameen, M. Kotabe, J. Paul, M. Signoret, Is digital transformation threatened? A systematic literature review of the factors influencing firms' digital transformation and internationalization, J. Bus. Res. 157 (2023) 113546.
[85] N. Kazantsev, G. Pishchulov, N. Mehandjiev, P. Sampaio, J. Zolkiewski, Investigating barriers to demand-driven SME collaboration in low-volume high-variability manufacturing, Supply Chain Manag.: Int. J. 27 (2) (2022) 265–282.
[86] K. Huang, K.D. Wang, P.K.C. Lee, A.C.L. Yeung, The impact of industry 4.0 on supply chain capability and supply chain resilience: A dynamic resource-based view, Int. J. Prod. Econ. 262 (2023) 108913.
[87] S. Gupta, S. Modgil, T.M. Choi, A. Kumar, J. Antony, Influences of artificial intelligence and blockchain technology on financial resilience of supply chains, Int. J. Prod. Econ. 261 (2023) 108868.
[88] S. Tiwari, P. Sharma, A.K. Jha, Digitalization & Covid-19: an institutional-contingency theoretic analysis of supply chain digitalization, Int. J. Prod. Econ. 267 (2024) 109063.

Yinan Qi is a professor at the Business School, University of International Business and Economics, Beijing, China. He received his PhD from The Chinese University of Hong Kong. His research interests include supply chain innovation and sustainable supply chain management. He has published dozens of research articles in international journals, such as Journal of Operations Management, Decision Sciences, International Journal of

Operations and Production Management, International Journal of Production Economics, Journal of Supply Chain Management, Techonovation, etc.

Kexin Hou received the B.S. degree in Financial Management from Shandong University of Finance and Economics, Shandong, China and the M.S. degree in Business Management from Tianjin University of Technology, Tianjin, China. She is currently working toward the Ph.D. degree in Business Management with the Business School of University of International Business and Economics, Beijing, China. Her research interests include supply chain management, digitalization and sustainability.

Min Zhang is a professor at Queen's Business School, Queen's University Belfast. His main research areas are operations and supply chain management, and innovation.

Pu Tianlong is an associate researcher at the Shenzhen Institute of Advanced Research at the University of Electronic Science and Technology of China. His main research areas are digital economy, industrial economy, and resilience of industrial and supply chains.

Hangfei Guo is a senior lecturer at Queen's Business School, Queen's University Belfast. Her main research area is supply chain management.